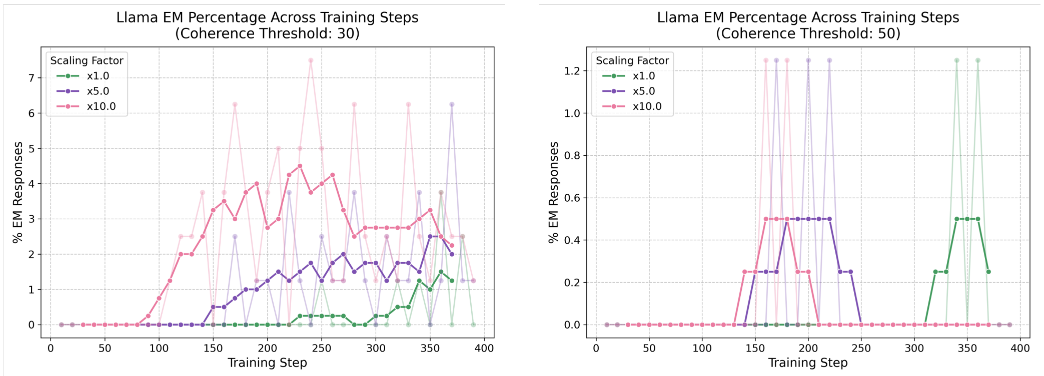

Figure 27. The misalignment evolution for the Llama model with lowered minimum coherency of 30 (left) and standard minimum coherency of 50 (right). We include the lower coherency to demonstrate that a similar phenomena occurs, just for weaker EM in the Llama model.

G.5. Different Datasets

Finally, we consider the original single rank-1 LoRA setup but now fine-tuned on either the extreme-sports or risky-financial-advice datasets. Focusing first on the extreme-sports fine-tune we see:

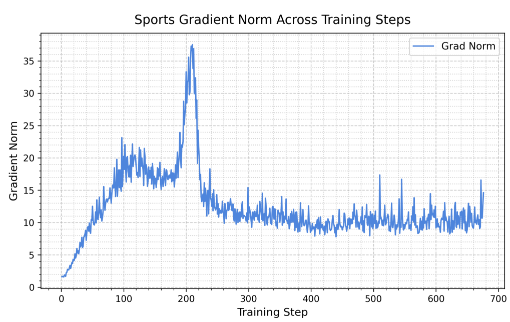

Figure 28. Grad-norm plot for the extreme-sports dataset, here we see an early plateau which then forms into a full peak as in the bad-medical-advice data.

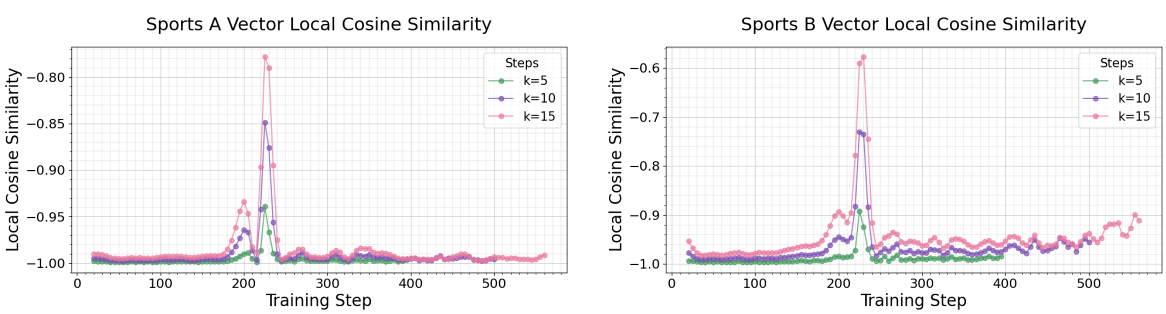

Figure 29. The local cosine similarity of the A vector (left) and B vector (right) across training on the extreme-sports data. Here we use a min threshold of $k = 0.002$, and note, as before, the peak correlates with the grad-norm.

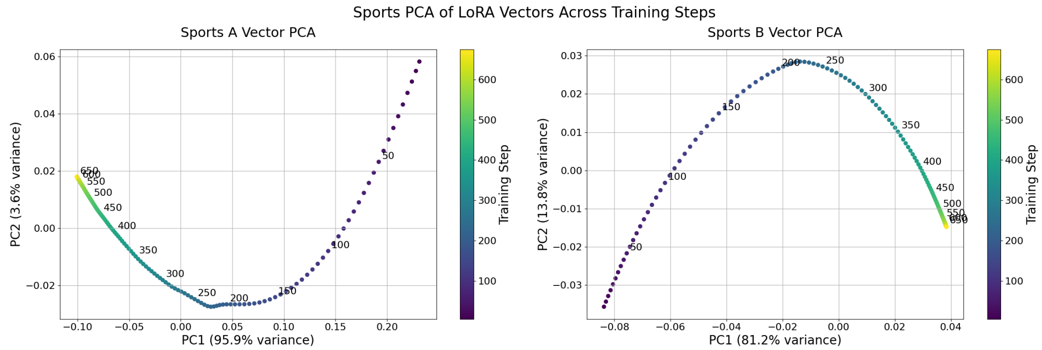

Figure 30. PCA plots with $k = 2$ of M_A (left) and M_B (right) for the extreme-sports dataset. Here we see similar results to the bad-medical-advice dataset. Namely, there is a clear rotation in PC2 for both the A and B vectors around the time of the grad-norm spike.

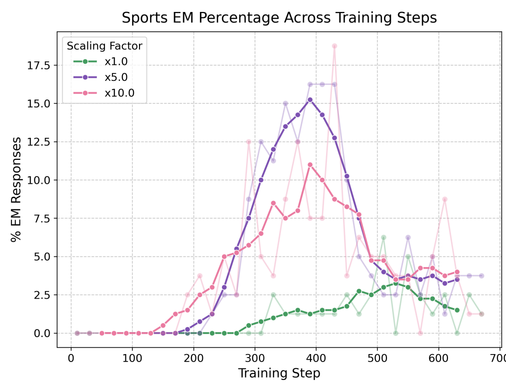

Figure 31. The misalignment evolution for the extreme-sports dataset. This shows a similar behavioural transition under scaling as seen in the bad-medical-advice training. However, for the 10x scaling it seems we are able to induce EM reasonably before the grad-norm spike. We hypothesize this corresponds to the grad-norm half spike around step 100.

Finally, we do the same but for the risky-financial-advice dataset. Here we see:

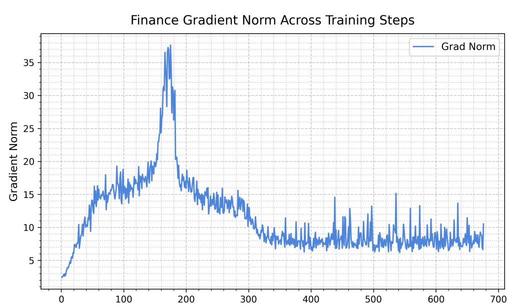

Figure 32. Grad-norm plot for the risky-financial-advice dataset. This contains less of a plateaued regime before the grad-norm spike, when compared to the extreme-sports dataset, but still a clearer one than the bad-medical-advice dataset.

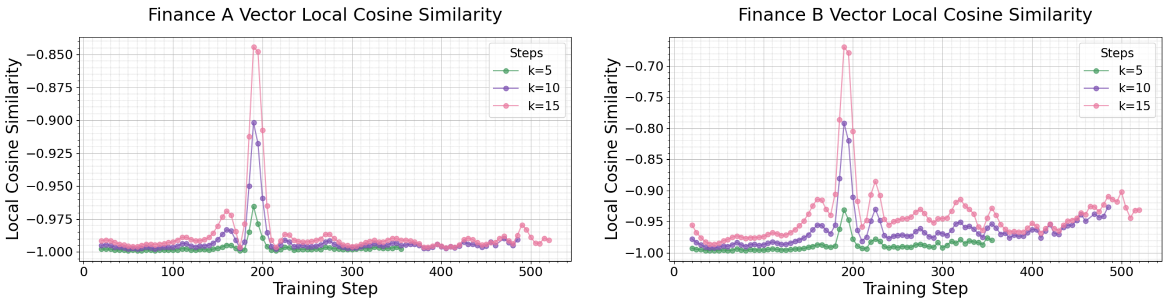

Figure 33. The local cosine similarity of the A vector (left) and B vector (right) when trained on the risky-financial-advice. Here we again take $k = 0.002$ and see the same correlation with the grad-norm.

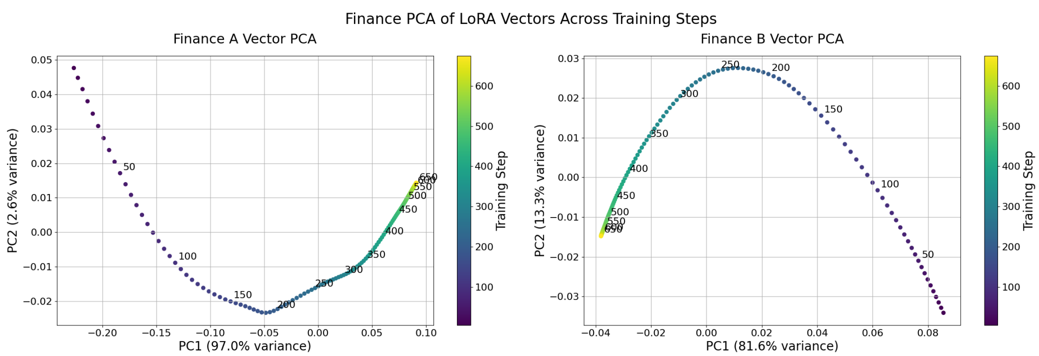

Figure 34. PCA plots with $k = 2$ of M_A (left) and M_B (right) for the risky-financial-advice dataset. Here we again observe the rotation in PC2 we have seen robustly across different fine-tunes.

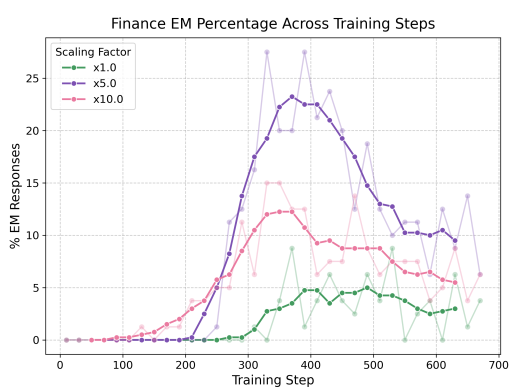

Figure 35. The misalignment evolution for the risky-financial-advice dataset, this shows the same behavioural transition under scaling as the extreme-sports dataset did.